

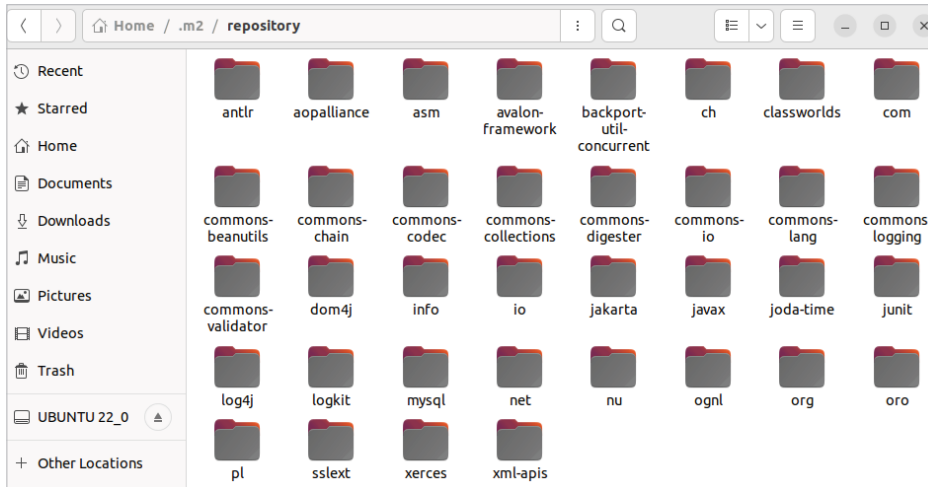


Figure 3: Folder view with all dependencies

Observe the logs of Maven command execution where it says *Building jar: /home/osfy/Desktop/spring-petclinic/target/spring-petclinic-2.7.3.jar*.

Now, we have a JAR file ready. The next step is to create a Docker image that contains the JAR file and runtime environment.

So, what is a Dockerfile? Well, it's nothing but the text document that provides build instructions to build the image with your application package.

```
FROM openjdk:11
MAINTAINER osfy
COPY target/spring-petclinic-2.7.3.jar app.jar
ENTRYPOINT ["java","-jar","/app.jar"]
```

Let's understand the above steps.

- **FROM** command: Uses the base image (openjdk:11) for subsequent instructions in Dockerfile.
- **COPY** command: Copies a file to the container.
- **ENTRYPOINT** command: Specifies entry point and manipulates the environment.

Place the Dockerfile in the *project root* directory. Now it is time to build a Docker image.

Building a Docker image and running a container

Execute the:

```
sudo docker image build -t petclinic:latest
```

...command from the *project root* directory. The output of the command will be similar to what's given below:

```
osfy@ubuntu-22-04-1-lts:~/Desktop/spring-petclinic$ sudo
docker image build -t petclinic:latest .
[sudo] password for osfy:
```

Sending build context to Docker daemon 66.47MB

```
Step 1/4 : FROM openjdk:11
11: Pulling from library/openjdk
001c52e26ad5: Pull complete
d9d4b9b6e964: Pull complete
2068746827ec: Pull complete
9dae329d350: Pull complete
d85151f15b66: Pull complete
66223a710990: Pull complete
db38d58ec8ab: Pull complete
Digest: sha256:99bac5bf83633e3c
7399aed725c8415e7b569b54e03e459
9e580fc9cdb7c21ab
Status: Downloaded newer image
```

```
for openjdk:11
--> 47a932d998b7
```

Step 2/4 : MAINTAINER osfy

```
--> Running in 782da3d9a384
Removing intermediate container 782da3d9a384
--> 8f17dd1a4292
```

Step 3/4 : COPY target/spring-petclinic-2.7.3.jar app.jar

```
--> e3ac33fc14f7
```

Step 4/4 : ENTRYPOINT ["java","-jar","/app.jar"]

```
--> Running in df55cf2cbbca
Removing intermediate container df55cf2cbbca
--> 28440427e82c
```

Successfully built 28440427e82c

Successfully tagged petclinic:latest

Now verify that the *ptclinic* image is available while you execute the *docker images* command:

```
osfy@ubuntu-22-04-1-lts:~/Desktop/spring-petclinic$ sudo
docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
petclinic	latest	28440427e82c	About a minute ago	707MB
nginx	latest	88736fe82739	2 weeks ago	142MB
gcr.io/k8s-minikube/kicbase	v0.0.35	7fb60d0ea30e	8 weeks ago	1.12GB
openjdk		11	47a932d998b7	4 months ago
			654MB	

Execute *docker run* command to create a container from the PetClinic image.

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ sudo docker run -d -p
8888:8080 petclinic:latest
b3915b15b7ea8fa94288ea132ec548dee0
488bed412ebe541c39ad13291ce157
```